

1(a)(i)	(approximately) 1 ; A value in 0.94 to 1 range to allow for mixed CHO and protein	1
1(a)(ii)	any 7 from: 1 <i>idea of</i> adding a fixed, number / mass, of peas to respirometer (below the shelf with soda lime) ; 2 <i>idea of</i> a control containing dead (sterilised) peas / glass beads of same mass ; 3 <i>idea of</i> inserting bung with tubing with clip and glass tubing with scale so that the apparatus is airtight / AW ; 4 <i>idea of</i> dipping glass tube into coloured liquid ; 5 <i>idea of</i> placing respirometer tube into a water-bath at a stated temperature in range 13°C to 18°C ; 6 <i>idea of</i> (opening clip and) leaving to equilibrate ; 7 <i>idea of</i> closing the clip and marking the position of dye <u>and</u> <i>idea of</i> starting stop-clock immediately ; 8 <i>idea of</i> measuring movement of dye on the scale, for 5 minutes / at timed intervals ; 9 <i>idea of</i> opening clip and resetting the coloured liquid ; 10 <i>idea of</i> doing minimum of three measurements in sequence <u>and</u> taking a mean ; 11 <i>idea of</i> removing carbon dioxide absorbent ; 12 <i>idea of</i> again measuring movement of dye on the scale, for 5 minutes / at timed intervals ; <i>must be linked to mp11</i> 13 low risk experiment or 14 medium risk experiment <u>and</u> carbon dioxide absorbent is harmful <u>and</u> wear gloves ; 15 AVP ;	7
1(b)	any 3 from: 1 find radius of glass tubing and use πr^2 (to find surface area) ; 2 multiply πr^2 by distance moved by liquid with carbon dioxide absorbent to find volume of oxygen used ; 3 multiply πr^2 by distance moved by liquid without carbon dioxide absorbent ; 4 add together the volume with carbon dioxide absorbent and volume without carbon dioxide absorbent (to find volume of carbon dioxide released) ; A marks on a correct formula e.g. oxygen consumed = $\pi r^2 \times$ distance moved with carbon dioxide absorbent = 2	3
1(c)(i)	1 <i>idea of</i> replacing the sodium hydroxide by (a named) oxygen absorbent / replacing air with nitrogen ; 2 <i>idea of</i> setting dye position, close to the tube containing peas / half way along the tube ; 3 look for liquid moving away from the peas (showing that carbon dioxide is produced in anaerobic conditions) ;	3
1(c)(ii)	trial 2 <u>and</u> the result, is anomalous / does not fit the pattern ;	1